A One Way Quantum Computer

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Abstract

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I. INTRODUCTION

I.1. Quantum Teleportation

In this section, I will give a brief summary of quantum teleportation. In the simplest case, quantum teleportation is realized on a 3-qubit state. Assume that Alice holds two qubit and Bob holds one, and Alice wants to teleport one of her qubit, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob.

First, Alice and Bob entangle their qubits (that is not the one waiting to be teleported) to a maximally entangled state. Then, the three qubit state becomes

$$|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle\right)$$

Then, Alice performs a unitary transformation on her two qubits $U = (H \otimes \mathbb{I})CNOT$, the resulting state is

$$\begin{aligned}|\Psi'\rangle = |00\rangle|\psi\rangle + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \\ + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)\end{aligned}$$

The input qubit has sucessfully transported to Bob's Hilbert space, with a distortion. However, this distortion can be corrected based on the measurement on Alice's Hilbert space.

- If Alice has measurement output $|00\rangle$, the qubit is not altered, hence we apply $\mathbb{I}$ on Bob's qubit.
- If Alice's measurement output is $|01\rangle$, the qubit experiences a bit flip, hence we apply a X gate on Bob's qubit.
- If Alice's measurement output is $|10\rangle$, the qubit experiences a π phase shift, so we apply a Z gate on Bob's qubit.
- If Alice's measurement output is $|11\rangle$, the qubit experiences both a bit flip and a π phase shift. Hence, we apply a ZX gate on Bob's qubit.

Combing these three steps, we sucessfully trasport the qubit from Alice's Hilbert space to Bob's Hilbert space without any error.

II. CLUSTER STATE

II.1. Definition of Cluster State

To allow the operation of quantum circuit on a lattice of qubits, a cluster state will be needed. Consider an n dimensional lattice of qubits. An n-qubit cluster $\mathcal{C}$ is a connected subset of the lattice containing n qubits. Then, the cluster state $|\phi_{\{\kappa\}}\rangle_{\mathcal{C}}$ is direct product of the n-qubit state following:

$$\begin{aligned}K^{(a)} &= \sigma_x^{(a)} \bigotimes_{b \in nbgh(a)} \sigma_z^{(b)} \\ K^{(a)}|\phi_{\{\kappa\}}\rangle_{\mathcal{C}} &= (-1)^{\kappa_a}|\phi_{\{\kappa\}}\rangle_{\mathcal{C}} \quad \forall a \in \mathcal{C}\end{aligned} \tag{1}$$

where $nbgh(a)$ is the neighborhood of a, namely the set of sites that are connected to site a, and $\kappa_a \in \{0,1\}$. Further, the set $\{\kappa_a | a \in \mathcal{C}\}$ specifies the cluster state $|\phi_{\{\kappa\}}\rangle_{\mathcal{C}}$. If two cluster states has κ, κ' different by at least one element, then the two cluster states are orthogonal to each other. Hence, the set of $\{|\phi_{\{\kappa\}}\rangle_{\mathcal{C}} | \{\kappa\} \in \{0,1\}^{|\mathcal{C}|}\}$ forms an orthogonal basis of the $2^{|\mathcal{C}|}$ dimensional Hilbert space.

II.2. Preparation

The first step of create a cluster state on a $\mathcal{C}_n$ cluster is to prepare a product state $|+\rangle_{\mathcal{C}} = \bigotimes_{a \in \mathcal{C}} |+\rangle_a$. Then, apply a unitary transformation

$$S^{(\mathcal{C})} = \prod_{a,b \in \mathcal{C} | b \in nbgh(a)} S^{ab} \tag{2}$$

$$S^{ab} = \frac{1}{2}\left(1 + \sigma_z^{(a)} + \sigma_z^{(b)} - \sigma_z^{(a)} \otimes \sigma_z^{(b)}\right)$$

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This unitary transformation impose a phase π on the state $|1\rangle_a \otimes |1\rangle_b$, and keeps the rest of the states $|0\rangle_a \otimes |0\rangle_b$, $|0\rangle_a \otimes |1\rangle_b$, $|1\rangle_a \otimes |0\rangle_b$ invariant.

Following this construction, we can write out the cluster state on 2, 3, 4 qubit cluster.

$$
\begin{aligned}
|\phi_{\{\kappa\}}\rangle_{\mathcal{C}2} &= \frac{1}{\sqrt{2}}\left(|0\rangle_1 |+\rangle_2 + |1\rangle_1 |-\rangle_2\right) \\
|\phi_{\{\kappa\}}\rangle_{\mathcal{C}3} &= \frac{1}{\sqrt{2}}\left(|+\rangle_1 |0\rangle_2 |+\rangle_3 + |-\rangle_1 |1\rangle_2 |-\rangle_3\right) \\
|\phi_{\{\kappa\}}\rangle_{\mathcal{C}4} &= \frac{1}{2}(|+\rangle_1 |0\rangle_2 |+\rangle_3 |0\rangle_4 + |+\rangle_1 |0\rangle_2 |-\rangle_3 |1\rangle_4 \\
&\quad + |-\rangle_1 |1\rangle_2 |-\rangle_3 |0\rangle_4 + |-\rangle_1 |1\rangle_2 |+\rangle_3 |1\rangle_4)
\end{aligned}
$$

III. CONSTRUCTION OF SINGLE QUANUTM STATE

III.1. CNOT

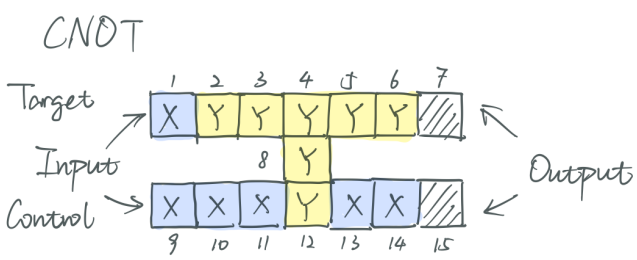

FIG. 1. Each box indicates a qubit site. The figure provides the measurement pattern on 15 qubit cluster to realize a CNOT gate.

In this construction, a CNOT gate is realized through a 15 qubit cluster, following the below steps:

- Prepare the initial state $|\psi_{in}\rangle_{1,9} \left(\bigotimes_{i\in\mathcal{C}_{15\setminus\{1,9\}}} |+\rangle_i\right)$
- Entangle the 15 qubit cluster with the unitary transformation $S^{(\mathcal{C}_{15})}$
- One qubit measurement is performed on every qubit in the cluster except for the output qubit(7, 15). The measurement pattern is visualized in Fig.1.
  - σ_x measurement: qubit 1, 9, 10, 11, 13, 14
  - σ_y measurement: qubit 2, 3, 4, 5, 6, 8, 12
  - The measured qubits would be destroyed and resulting gate is $U'_{CNOT} = U_{\Sigma,CNOT} CNOT(c,t)$
  - $U_{\Sigma,CNOT} = {\sigma_x^{(c)}}^{\gamma_x^{(c)}} {\sigma_x^{(t)}}^{\gamma_x^{(t)}} {\sigma_z^{(c)}}^{\gamma_z^{(c)}} {\sigma_z^{(t)}}^{\gamma_z^{(t)}}$ with

$$
\begin{aligned}
\gamma_x^{(c)} &= s_2 + s_3 + s_5 + s_6 \\
\gamma_x^{(t)} &= s_2 + s_3 + s_8 + s_{10} + s_{12} + s_{14} \\
\gamma_z^{(c)} &= s_1 + s_3 + s_4 + s_5 + s_8 + s_9 + s_{11} + 1 \\
\gamma_z^{(t)} &= s_9 + s_{11} + s_{13}
\end{aligned}
$$

III.2. $SU(2)$ Rotation

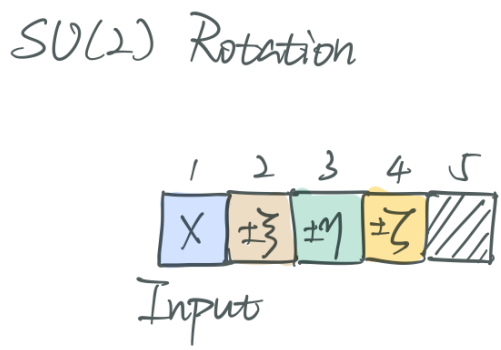

FIG. 2. The figure provides the measurement pattern on 5 qubit cluster to realize an arbitrary SU(2) rotation.

Similarly, a general $SU(2)$ rotation can be realized on a 5 qubit cluster. For any $U \in SU(2)$ on a qubit, it can be decomposed into the form $U = U_X[\zeta]U_Z[\eta]U_X[\xi]$, correspond to Euler rotation and its three Euler angles. The procedure is as follow:

- Prepare the state $|\Psi\rangle_{\mathcal{C}_5} = |\psi_{in}\rangle_1 \left(\bigotimes_{i=2}^{5} |+\rangle_i\right)$
- Create the cluster state by performing the unitary operation $S^{(\mathcal{C}_5)}$
- Measure the first 4 qubits in the following order. Each measurement correspond to a step of Euler rotation
  - Measure qubit 1 in $\mathcal{B}_1(0)$.
  - Measure qubit 2 in $\mathcal{B}_2(-\xi(-1)^{s_1})$.
  - Measure qubit 3 in $\mathcal{B}_3(-\eta(-1)^{s_2})$.
  - Measure qubit 4 in $\mathcal{B}_4(-\zeta(-1)^{s_2+s_3})$.

  With $\mathcal{B}_j(\varphi_j) = \left\{\frac{|0\rangle_i + e^{i\varphi}|1\rangle_j}{\sqrt{2}}, \frac{|0\rangle_i - e^{i\varphi}|1\rangle_j}{\sqrt{2}}\right\}$
- Then, we have performed a rotation $U'_{Rot} = U_{\Sigma,Rot} U_{Rot}[\eta,\xi,\zeta]$, with the byproduct operator $U_{\Sigma,Rot} = \sigma_x^{s_2+s_4}\sigma_z^{s_1+s_3}$

III.3. Hadamard and $\pi/2$-Phase Gate

In the previous section, I described the scheme to construct the measurement pattern for a general $SU(2)$ rotation. The measurement must be performed in the order

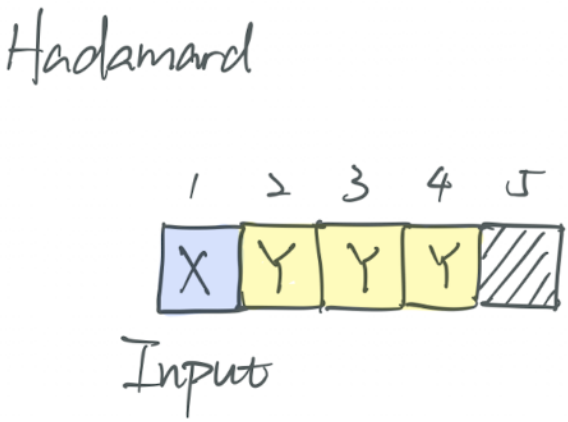

FIG. 3. The figure provides the measurement pattern on 5 qubit cluster to realize an arbitrary Hadamard Gate.

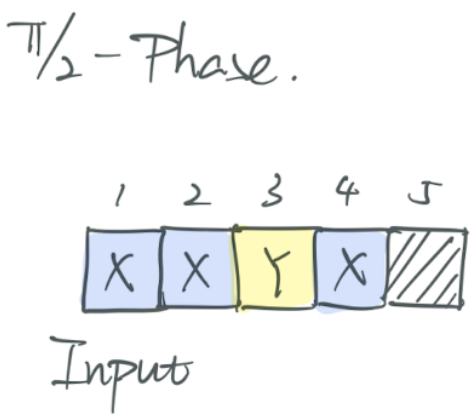

FIG. 4. The figure provides the measurement pattern on 5 qubit cluster to realize an arbitrary $\pi/2$-phase gate.

specified. However, there is a special class of single qubit gates where the measurement can be performed simultaneously. In this section, I will give two examples of such gates, Hadamard gate and $\pi/2$-phase gate.

Both gates are constructed on a 5-qubit cluster.

- Prepare the 5-qubit cluster state
- Measurement of the first 4 qubit can be performed simultaneously
  - Hadamard Gate: $\sigma_x^{(1)}$, $\sigma_y^{(2)}$, $\sigma_y^{(3)}$, $\sigma_y^{(4)}$, $U_{\Sigma,H} = \sigma_x^{s_1+s_3+s_4}\sigma_z^{s_2+s_3}$
  - $\pi/2$-phase Gate: $\sigma_x^{(1)}$, $\sigma_x^{(2)}$, $\sigma_y^{(3)}$, $\sigma_x^{(4)}$, $U_{\Sigma,H} = \sigma_x^{s_2+s_4}\sigma_z^{s_1+s_2+s_3+1}$.

IV. QUANTUM CIRCUIT: LOGICAL NETWORK

In this section, I will show how the measurement pattern indeed realize a quantum circuit and is equivalent as a chain a quantum gates. Namely, I will be showing the below two schemes are equivalent.

For any form of procedure, the cluster needed $\mathcal{C}(g)$ can be decomposed into three parts, an input cluster $\mathcal{C}_I(g)$, a body cluster $\mathcal{C}_M(g)$, and an output cluster $\mathcal{C}_O(g)$. with the properties

$$\begin{aligned} C_I(g) \cup C_M(g) \cup C_O(g) &= C(g) \\ C_I(g) \cap C_M(g) &= \emptyset \\ C_I(g) \cap C_O(g) &= \emptyset \\ C_M(g) \cap C_O(g) &= \emptyset \end{aligned}$$

Scheme 1 (Quantum Gates):

- Start with the state $|\psi_{in}\rangle_{1,9} \left(\bigotimes_{i\in\mathcal{C}_M\cup\mathcal{C}_O} |+\rangle_i\right)$
- Prepare the cluster state by applying $S^{(\mathcal{C}(g))}$
- Measure the qubits in the cluster $\mathcal{C}_I(g)\cup\mathcal{C}_M(g)$ with some projector

$$P^{(\mathcal{C}_I(g)\cup\mathcal{C}_M(g))} = \bigotimes_{k\in\mathcal{C}_I(g)\cup\mathcal{C}_M(g)} \frac{1+(-1)^{s_k}\vec{r_k}\cdot\vec{\sigma}_k}{2}$$

  where s_k is some binary value based on the previous measurement outcomes, and $\vec{r_k}$ is a unit vector in the x-y plane.

Then, the measurement output is

$$\begin{aligned} |\Psi\rangle &= P^{(\mathcal{C}_I(g)\cup\mathcal{C}_M(g))} S^{(\mathcal{C}(g))} |\psi_{in}\rangle_{1,9} \left(\bigotimes_{k\in\mathcal{C}_M\cup\mathcal{C}_O} |+\rangle_i\right) \\ &= \left(\bigotimes_{k\in\mathcal{C}_I(g)\cup\mathcal{C}_M} |s_k\rangle_{k,\vec{r_k}}\right) \otimes |\psi_{out}\rangle \\ &= \left(\bigotimes_{k\in\mathcal{C}_I(g)\cup\mathcal{C}_M} |s_k\rangle_{k,\vec{r_k}}\right) \otimes U_{\Sigma,g} U_g |\psi_{in}\rangle \end{aligned}$$

with $U_{\Sigma,g} = \bigotimes_{i=1}^{n} \left(\sigma_x^i\right)^{x_i} \left(\sigma_z^i\right)^{z_i}$. $\sigma_{x/z}^i$ denotes pauli matrices acting on the i-th qubit in the output cluster, and x_i and z_i are binary values based on the previous measurement outcomes.

V. CONCLUSION

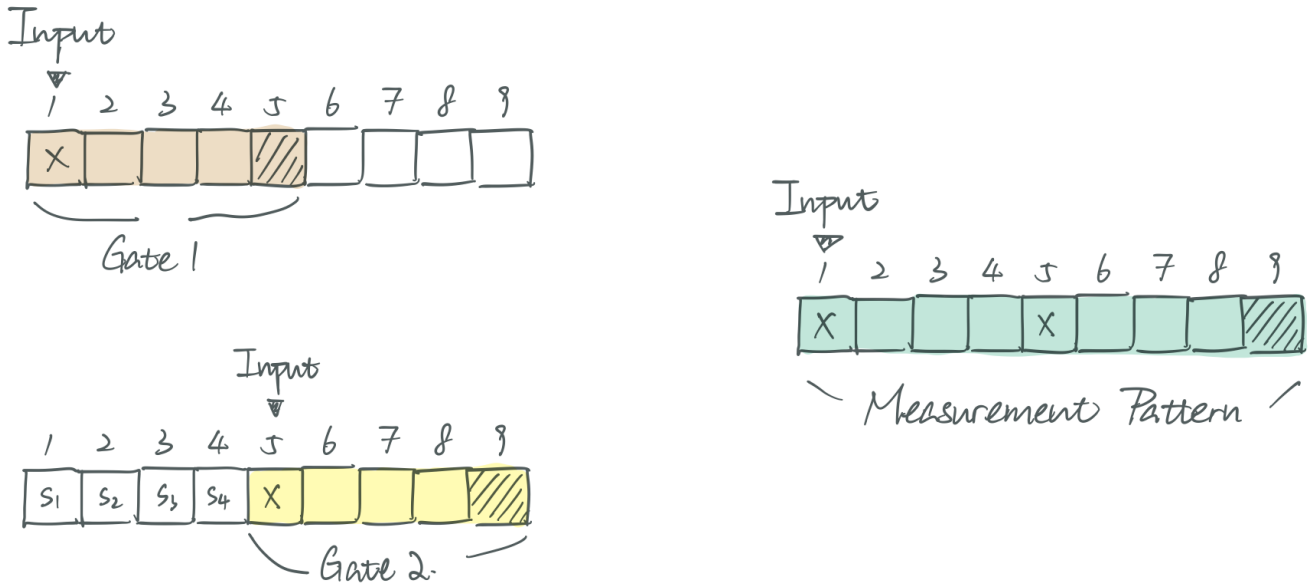
Input
1 2 3 4 5 6 7 8 9
X
Gate 1
Input
1 2 3 4 5 6 7 8 9
S1 S2 S3 S4 X
Gate 2
Input
1 2 3 4 5 6 7 8 9
X X
Measurement Pattern

FIG. 5.